Ahmed Samak

Senior Software Engineer and Team Lead | Full-Stack, Cloud, Web Graphics and Data Visualization
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SUMMARY

Senior Software Engineer and Team Lead with 6+ years of experience building web platforms, high-performance desktop software, and interactive 2D/3D graphics. Team Lead on Llumen, a decision intelligence and data visualization platform for government leadership in the UAE, owning the product's map and 2D/3D graphics, and Forward Deployed Engineer delivering it on Microsoft Azure and air-gapped on-premise infrastructure with Kubernetes and Helm. Hands-on with TypeScript, Go, Python, Three.js, and microservices, with a background in Modern C++ for FDA-approved medical software and in freelance web game development.

EXPERIENCE

Team Lead / Senior Software Engineer Feb 2026 – Present
Pixonal | Sheikh Zayed, Egypt - Dubai, UAE (Hybrid)

- Lead an engineering team working on Llumen, Pixonal's decision intelligence platform that brings data, analytics, models, and AI into one environment for government leaders; own sprint planning, task breakdown, code reviews, mentoring, and hiring.
- Own the graphics layer of Llumen: develop, maintain, and optimize all map visualizations and the 2D and 3D graphics used to visualize data.
- Re-engineered map rendering with GPU instancing and custom shaders, making it more than 100x faster: the map now renders over 1 million markers smoothly, where it previously struggled at 100K.
- Deliver bi-weekly releases of a large microservices platform (Go, TypeScript, Python, PostgreSQL), coordinating scope, QA, and rollout across cloud and client-hosted environments.
- Act as Forward Deployed Engineer for UAE government entities: run on-site meetings and requirement workshops, translate operational needs into product features and configurations, and help clients get the most out of the platform.
- Deploy and operate the platform on Microsoft Azure VMs with Kubernetes and Helm charts, and on air-gapped on-premise servers; travel to client sites to resolve deployment and network connectivity issues.
- Administer Azure resources for client and internal environments, including virtual machines, AKS, virtual networks, and storage.

Senior Full-Stack Engineer Jan 2025 – Feb 2026
Pixonal | Sheikh Zayed, Egypt - Dubai, UAE (Hybrid)

- Built immersive, data-driven products and dashboards for leaders and decision-makers in government.
- Developed Go and Python microservices for data ingestion, management, and analytics, including JSON-to-Parquet conversion pipelines, data summarization, and AI/LLM-based analysis.
- Designed an optimized stories, whiteboards, and dashboards system with complex components and interactive map visualizations.
- Built a scalable 3D GeoJSON visualization library on Mapbox and Three.js.
- Optimized data loading and rendering so graphics-heavy dashboards run smoothly in the browser for many concurrent users.
- Tools: Go, TypeScript, Python, FastAPI, C# (.NET), PostgreSQL, Supabase, React, Next.js, Mapbox, Three.js, Docker, Kubernetes.

Senior Software Engineer Apr 2022 – Dec 2024
MedSoft | Cairo, Egypt (Hybrid)

- Worked on PPDS, FDA-approved high-performance medical software for simulating orthognathic surgery (now owned by Johnson & Johnson): six plugins covering the procedure steps, letting surgeons simulate the operation, generate surgical guides, plates, and screws, and review results through a web plugin.
- Re-designed the screw placement and editing feature used across plugins into a reusable widget, saving about 20% of development time in following sprints; contributed to system design and sprint planning for many features.
- Developed bridge classes connecting Qt/QML user interface components to the C++ back-end.
- Implemented data-model algorithms to serialize and deserialize patient data into case folders, and features to design and generate 3D surgical hardware.
- Collaborated with teams in the USA and India using Scrum and Jira in an Agile environment.
- Tools: Modern C++17, VTK, Qt/QML, CMake, Software Design Patterns.

Senior Front-End Software Engineer Sep 2022 – Feb 2023
Bld.ai | USA (Remote)

- Built responsive, dynamic user interfaces for web and mobile platforms using React and React Native.
- Worked with design and back-end teams to integrate APIs and deliver seamless user experiences.

Unreal Engine Developer Aug 2021 – May 2022
Archrete | New Cairo, Egypt (On-Site)

- Worked on iMaket, a real-time simulation product for major architectural projects in the Middle East.
- Designed and implemented Unreal Engine 5 plugins in C++ exposed as Blueprints, extending simulation functionality and unblocking projects.
- Worked with clients to understand requirements and delivered tailored .NET solutions in C#.

JavaScript Game Developer Aug 2020 – Aug 2021
ElMoasserBooks | Cairo, Egypt (On-Site)

- Developed educational web games with designers and writers, optimized to run smoothly on phones and laptops.
- Used JavaScript, CreateJS, Three.js, Adobe Animate, and Unity.

Freelance Interactive Web and Game Developer 2020 – Present
Self-Employed | Remote

- Build animated 2D web games for educational and commercial clients, optimized to run smoothly on phones, tablets, and desktops.
- Develop lightweight real estate pitching simulations: interactive Three.js and 2D web apps that let sales teams present properties on any device.
- Built Reveola, a Facebook Instant Game in React and the Instant Games API, a gamified version of the Reveola review app for movies and places.
- Tools: JavaScript, TypeScript, Three.js, Tween.js, CreateJS, HTML5 Canvas, React.

SKILLS

Languages: TypeScript, JavaScript, Go, Python, C++, C, C#, SQL, PL/pgSQL, Java, QML, HTML, CSS
Backend and Data: Microservices, REST APIs, FastAPI, .NET, PostgreSQL, Supabase, Data Pipelines, Parquet, JSON, GeoJSON
Cloud and DevOps: Microsoft Azure (VMs, AKS, Networking, Storage), Kubernetes, Helm, Docker, Linux, On-Premise and Air-Gapped Deployments, Git, GitHub, GitLab, Bitbucket
Frontend and Visualization: React, Next.js, React Native, Redux, Mapbox, Data Visualization, Dashboards
Web Graphics and Games: Three.js, WebGL, Shaders, GPU Instancing, Tween.js, CreateJS, HTML5 Canvas, 2D Web Games, Unity, Unreal Engine 5
Systems and Desktop: Modern C++, STL, Qt/QML, VTK, CMake
AI: LLMs, Model Context Protocol (MCP), AI-Assisted Development (Claude Code, Cursor)
Leadership and Practices: Team Leadership, Mentoring, Code Review, Hiring, Sprint Planning, Agile and Scrum, Client Requirements, Stakeholder Management, System Design, Design Patterns, SOLID, OOP, Jira

EDUCATION AND CERTIFICATIONS

Microsoft Certified: Azure Administrator Associate (AZ-104), Microsoft, Expected 2026
B.Sc. in Computer Science, Menofia University, Egypt, Sep 2015 – Jul 2019
TOEFL iBT, Score 95, ETS, Jan 2020
ITIDA Next Technology Leaders, 5 Courses, Smart Village, Giza, Egypt
Game Design and Development Specialization, Michigan State University (Coursera), May 2018

PROJECTS

- **Solar System Simulation**: Unity 3D simulation with relatively accurate planet distances, camera focus on any body, and adjustable orbit speeds.
- **Pixie** (Graduation project, team lead): Android app in Java that scans paper and reads text aloud through voice commands to help visually impaired people.

ACTIVITIES

- Co-Founder and Practice Sessions Instructor, ICPC Menofia University Team (2017 – 2019); ICPC Arab Collegiate Programming Contest contestant (2018).
- Event Coordinator, MUFIX Student Activity (2017 – 2019); Executive Member, Microsoft Student Partners Tech Club (2015 – 2017).

LANGUAGES

Arabic (Native), English (Fluent), German (Beginner)